Lecture 15-17 Scribe

L15_L17.pdf
PDF

Generate a lecture scribe suitable for exam preparation from the provided lecture 15-17. While making this, reason step by step using definitions, assumptions, derivation, proof, and example solution exactly as given in the slides. Include the step by step reasoning so that when we revise for our exams it becomes easy. All explanations and reasoning must be from the provided ppt. do not introduce any new examples or theory apart from what is there in the ppt. Explain every definition and assumption explicitly. derive the formulas and solve the examples step by step using whatever is there in the slide. The tone should be formal and academically suitable for revision. try to include every example and question from the pdf

I carefully used the uploaded slides **L15–L17** only and converted them into an **exam-oriented lecture scribe** with definitions, assumptions, derivations, and solved examples exactly following the PPT content.
Source:

Lecture 15–17 Scribe

Joint Random Variables and Joint Distributions

Course: CSE400 – Fundamentals of Probability in Computing
Instructor: Dhaval Patel, PhD
Date: March 10, 2026

---

1. Why Study Joint Random Variables?

In many scientific and real-world situations, we are interested in **two or more random variables simultaneously** rather than a single random variable.

The slides explain that studying variables together allows us to analyze the **relationship between them**.

Examples

The lecture provides several motivating examples:

1. **Height and weight of giraffes**
2. **IQ and birthweight of children**
3. **Frequency of exercise and heart disease rate**
4. **Air pollution level and respiratory illness rate**
5. **Number of Facebook friends and age of members**

These examples illustrate that **multiple variables may influence each other**.

Real System Examples

Smart Transportation

Two random variables are defined:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question posed in the slide:

> Can speed be high when traffic density is also high?

This question illustrates the **relationship between variables**.

Applications mentioned:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Wireless Network Performance

Two variables:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

The slides ask:

> If SNR is high, will throughput always be high?

Reason: Network congestion and interference can affect throughput even if SNR is high.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Other Examples Mentioned

- Weather prediction
  $X =$ Rainfall, $Y =$ Temperature
- Drone delivery
  $X =$ Wind Speed, $Y =$ Delivery Time
- Rolling two dice
  X and Y are the values of the dice.

Key Idea

In such situations, the random variables have a **joint distribution**, which allows us to:

1. Compute probabilities involving **both variables**
2. Understand the **relationship between variables**

If variables are not independent, we analyze their relationship using:

- **Covariance**
- **Correlation**

---

2. Discrete Case – Joint PMF

Setup

Suppose X and Y are **discrete random variables**.

$$X = \{x_1, x_2, ..., x_n\}$$

$$Y = \{y_1, y_2, ..., y_m\}$$

The pair (X, Y) takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), ..., (x_n, y_m)\}$$

This means every possible combination of values of X and Y is considered.

Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This gives the **probability that both events occur simultaneously**.

---

3. Joint PMF Table Representation

The joint PMF can be organized in a **table format**.

X\Y	y_1	y_2	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_m)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_m)$
...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_m)$

Key Properties

1. $$0 \leq p(x_i, y_j) \leq 1$$

2. $$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

This means the **total probability over all possible pairs equals 1**.

---

4. Example – Rolling Two Dice

Experiment

Roll **two fair dice**.

Define random variables:

$$X = \text{value on the first die}$$

$$T = \text{total sum of both dice}$$

Each outcome (i, j) has probability:

$$\frac{1}{36}$$

because there are **36 equally likely outcomes**.

Joint Probability Table

The table in the slide shows probabilities of (X, T).

Entries are either:

$$0 \quad \text{or} \quad \frac{1}{36}$$

Observation from slide:

- Entries are **0 or** $1/36$
- X and T are **not independent**

---

5. Continuous Case – Joint PDF

The lecture explains that the continuous case is analogous to the discrete case.

Discrete Case	Continuous Case
Discrete values	Continuous intervals
Joint PMF	Joint PDF
Sums	Double integrals

Domain of Variables

If

$$X \in [a, b]$$

$$Y \in [c, d]$$

then the pair (X, Y) takes values in the **product region**

$$[a, b] \times [c, d]$$

Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** if:

$$P((X, Y) \text{ lies in small rectangle } dx\, dy) \approx f(x, y)\, dx\, dy$$

Interpretation:

Probability of a small region ≈ density × area.

Key Properties

1.

$$f(x, y) \geq 0$$

2.

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Note from the slide:

> The joint PDF may take values greater than 1 because it is a density, not a probability.

---

6. Continuous Joint PDF Example

Given

$$f(x, y) = 4xy$$

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

The pair (X, Y) lies in the **unit square**

$$[0, 1]^2$$

Task 1 – Show that $f(x, y)$ is a valid joint PDF

We check the **normalization condition**.

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step 1: Integrate with respect to x

$$\int_0^1 (4xy)dx = 4y \int_0^1 xdx$$

$$= 4y \left[\frac{x^2}{2} \right]_0^1$$

$$= 2y$$

Step 2: Integrate with respect to y

$$\int_0^1 2ydy$$

$$= [y^2]_0^1$$

$$= 1$$

Thus:

$$\int_0^1 \int_0^1 4xydxdy = 1$$

Therefore:

- $f(x, y) \geq 0$
- Total probability = 1

Hence it is a **valid joint PDF**.

Task 2 – Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

The region corresponds to:

$$0 \leq x \leq 0.5$$

$$0.5 \leq y \leq 1$$

Step 1: Write the probability

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Step 2: Integrate with respect to y

$$\int_{0.5}^{1} 4xy \, dy$$

$$= 4x \left[\frac{y^2}{2} \right]_{0.5}^{1}$$

$$= 2x(1 - 0.25)$$

$$= 2x(0.75)$$

$$= 1.5x$$

Step 3: Integrate with respect to x

$$\int_0^{0.5} 1.5x \, dx$$

$$= 1.5 \left[\frac{x^2}{2} \right]_0^{0.5}$$

$$= 1.5(0.125)$$

$$= \frac{3}{16}$$

Thus

$$P(A) = \frac{3}{16}$$

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation used in the slides:

$$X \leq x, Y \leq y$$

This represents the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

The **joint cumulative distribution function** is defined as

$$F(x, y) = P(X \leq x, Y \leq y)$$

---

8. Joint CDF – Continuous Case

If X and Y have joint density $f(x, y)$ over $[a, b] \times [c, d]$, then

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering the Joint PDF

Because there are two variables, we use **partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

---

9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation:

To compute $F(x, y)$:

1. Add probabilities from rows where $x_i \leq x$

2. Add probabilities from columns where $y_j \leq y$

---

10. Properties of the Joint CDF

The joint CDF $F(x, y)$ must satisfy several properties.

Property 1

$$F(x, y) \text{ is non-decreasing}$$

If x or y increases, $F(x, y)$ must stay constant or increase.

Property 2

$$F(x, y) = 0$$

at the **lower-left corner of the joint range**.

If the corner is

$$(-\infty, -\infty)$$

then

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

Property 3

$$F(x, y) = 1$$

at the **upper-right corner** of the joint range.

If the corner is

$$(\infty, \infty)$$

then

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

Quick Exam Revision Summary

Important formulas to remember:

Joint PMF

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

PMF Property

$$\sum_i \sum_j p(x_i, y_j) = 1$$

Joint PDF

$$P((X, Y) \in region) = \iint_{region} f(x, y)\, dx\, dy$$

Normalization

$$\iint f(x, y) dx dy = 1$$

Joint CDF

$$F(x, y) = P(X \leq x, Y \leq y)$$

Continuous Case

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recover PDF

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

If you want, I can also make a **perfect LaTeX lecture scribe (same format professors expect for CSE400 submissions)** so you can directly submit or revise before exams. That will save you a lot of time. 🚀